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W. O. BENNETT, JR

2,391,816

HAIRSPRING COLLET

Filed July 29, 1944

FIG. 1.

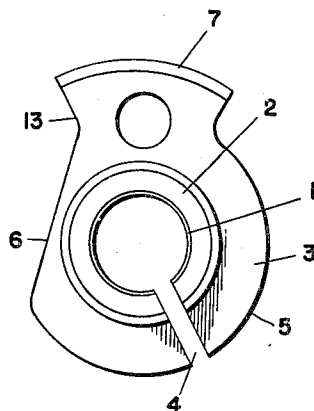


FIG. 2.

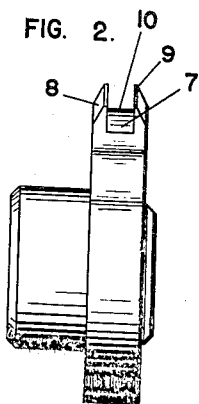


FIG. 3.

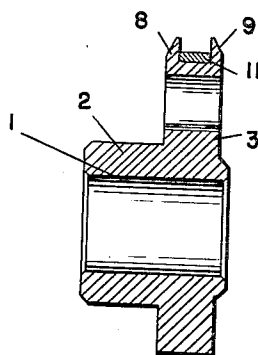


FIG. 4.

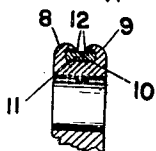
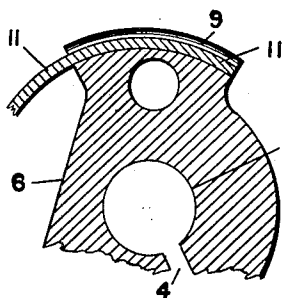


FIG. 5.



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HAIRSPRING COLLET

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7 Claims. (Cl. 58—115)

This invention is directed to an improvement in hairspring collets, more particularly to that type of collet which is counterpoised to balance out of poise effect of the hairspring.

This invention is an improvement on the collet which forms part of the patent of G. P. Luckey #2,037,729 and enhances that patent in certain details.

The object of the present invention is to provide a collet which is so shaped and designed as to counterbalance out of poise effect of the hairspring.

It is the object of the present invention to provide a particular means of securing the end of the hairspring to the collet, which securing is accomplished without in any way distorting the hairspring and having the clamping position such that the hairspring flexes from a single plane normal to the hairspring blade.

It is a further object of the present invention to provide a hairspring collet having an extended hub to provide a firm and secure bearing on the balance staff to prevent canting of the collet on the staff and eliminate the subsequent necessary adjustment.

A still further object of the present invention is to provide a collet formed with a calculated extended groove which coincides with the spiral of the hairspring, eliminating the necessity for truing the hairspring after assembly and making it unnecessary to do any bending or straightening of the spring after it is secured in position.

A further object of the present invention is to provide means for clamping the end of the hairspring which is not concentric with the center of the collet and which securely connects the spiral of the hairspring with the balance staff.

A still further object of the present invention is to provide a collet of a shape and design so that position error will be eliminated and regardless of the position in which the watch is maintained the effective balance of the assembled collet and hairspring is perfect.

It is a still further object of the present invention to provide a peripheral clamping groove which is in the form of a spiral to receive the free end of the hairspring and to clamp the hairspring in said groove with a clamping action which is normal to the flat surface of the hairspring.

The invention is illustrated in the accompanying drawing in which:

Fig. 1 is a plan view of the collet without the hairspring.

Fig. 2 is a side elevation of the same.

Fig. 3 is a cross-section of the collet showing a portion of the hairspring in position but not clamped.

Fig. 4 is a detailed sectional view showing a portion of the hairspring in clamped position and

Fig. 5 is a detailed sectional view further illustrating a portion of the hairspring in clamped position.

The hairspring collet forming the subject of this invention is formed with an interior bearing surface 1 adapted to fit over and to be held in tight relationship to a balance staff (not shown).

Exterior of the bearing surface is an extended hub 2 and a counterpoising flange 3, while a groove 4 is cut through both the counterpoising flange and the hub to provide the necessary resiliency to secure the hairspring collet on the balance staff.

The counterpoising flange is formed with a concentric portion 5, a flat portion 6 and a spiral clamping peripheral groove 7. The groove 7 has upstanding sides 8 and 9 and a flat hairspring receiving section 10, the entire groove being of a curve which is not concentric with the center of the collet and which is formed to correspond to the spiral of the hairspring 11. The collet is attached to the hairspring as shown in Figs. 4 and 5, the spring 11 lying on the flat surface 10 of the groove 7 and held in position by rolling or bending the sides 8 and 9 over so that the extreme ends of these sides engage the top surface of the hairspring at 12 producing a downward pressure so that the hairspring is secured by pressure exerted against the flat surface 10 of the groove 7. This prevents distortion of the spring by any stresses or strains which are generally set up where the clamping pressure is directed against the edges of the spring rather than the flat surfaces.

In assembling the hairspring and collet and balance staff the usual practice is to pin the hairspring to the collet with a tapered pin which is driven, together with the hairspring in a round hole, which produces a cross curvature of the hairspring and also distorts the hairspring from the true spiral curve with which it is formed. With the collet which forms the subject of the present invention there is no distortion of the hairspring. Because of the curvature of the groove 7 and the angled portion 13 of the collet the point of flexure of the hairspring is in a single plane normal to the flat surface of said hairspring which is not possible with the type of collet now in use. The clamping of the hairspring without distortion and the point of flexure being

in a single plane allows the hairspring to flex equally in either direction and provides an equal restraining force to the balance wheel. While the difference is extremely small, this difference, when multiplied by the many thousands of times a watch ticks in 24 hours, is sufficient to cause an appreciable error in the period of the balance wheel.

The extended hub 2 of the improved collet provides a greater bearing surface on the balance staff and prevents the canting or tilting of the collet on the balance staff during assembly and subsequent adjustment for beat position of the balance wheel which partially eliminates the need of the operation known as truing of the hairspring. It is the custom now to overcome this canting together with the stresses and strains set up by improper clamping of the hairspring, to cold work the hairspring by the use of tweezers and mainly by long experience and practice to bring the center of the hairspring spiral into coincidence with the center of the balance staff, and set the plane of the hairspring normal to said balance staff. With the improved collet the hairspring remains in a true spiral, thus eliminating the costly and delicate operation of truing. The operation of truing is carried out by cold working the hairspring and contributes to a drifting rate in the timekeeping of the instrument.

It has been proved through experiment that the construction of this improved collet has eliminated errors in the timekeeping quality and has eliminated the necessity of truing the hairspring together with the consequent cold working of the spring which even if carried out to perfection in time causes error in the period of the balance wheel.

What is claimed is:

I claim:

1. A hairspring collet comprising a hub, a counterpoised flange having a hairspring receiving groove formed with a spiral seat conforming to and matching the hairspring fitting therein, said groove and seat not being concentric with said hub.

2. A hairspring collet comprising a hub and a counterpoised flange, said flange having a spirally formed peripheral hairspring receiving groove formed with a spiral seat conforming to and matching the hairspring fitting therein.

3. A hairspring collet made up of a hub and a counterpoised flange, said flange being formed with a spiral peripheral hairspring receiving groove formed with a spiral seat conforming to and matching the hairspring fitting therein, said groove having sides extending therefrom and adapted to be intumed to secure the hairspring against the seat and within the groove.

4. A hairspring collet comprising a hub and a counterpoised flange, said counterpoised flange being formed with a peripheral portion concentric with said hub, a flat cut away portion and a spiral peripheral hairspring receiving groove formed with a spiral seat conforming to and matching the hairspring fitting therein and means associated with said groove for securing the hairspring against the seat and within the groove.

5. A hairspring collet comprising an extended hub and a counterpoised flange, said flange being formed with a spiral peripheral hairspring receiving groove having a spiral seat conforming to and matching the hairspring fitting therein, the termination edge of said groove being at such an angle to the center of said flange as to present a plain surface normal to a hairspring secured in said groove.

6. A hairspring collet forming part of an effectively balanced hairspring assembly comprising a hub portion and a flange portion, said flange portion having a part formed with a peripheral spiral hairspring receiving groove, having a spiral seat conforming to and matching the hairspring fitting therein and means for securing said hairspring in said groove by pressure exerted normally to the flat surface of said hairspring, the juncture of said hairspring and said groove being in a plane perpendicular to said flat surface of said hairspring to provide equal flexure of said hairspring in either direction from said juncture.

7. A hairspring collet forming part of an effectively balanced hairspring assembly comprising a hub portion and a flange portion, said flange portion being formed with a concentric part, a flat cutaway part, and a spiral peripheral hairspring receiving groove having a spiral seat conforming to and matching the hairspring fitting therein and means integral with said flange portion for securing said hairspring against said seat and in said groove.

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